

Fardeen Bablu

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EDUCATION

Vanderbilt University, Nashville, TN

May 2026

Bachelor of Science in Computer Science, Mathematics, Public Health

Activities: Head Resident Advisor, Entrepreneurship Society, First Generation VU, Assoc. of Bengali Students

SKILLS

Programming: Swift, TypeScript/JavaScript, Python, C++, Rust, SQL

Libraries: SwiftUI, React Native, Expo, React, Node.js, Firebase, PyTorch, Svelte, Tauri, Three.js

Tools: Git, Stripe, Plaid, CI/CD, AWS, GCP, Docker, Figma

EXPERIENCE

Founder & iOS Engineer

Remote

Niyah, a focus app where users stake money on work sessions niyah.live

Jan. 2026 – Present

- Shipped a real-money focus app whose native Swift Screen Time extensions earned Apple's FamilyControls Distribution.
- Built a custom ManagedSettings shield, WidgetKit widget, and ActivityKit Live Activity in SwiftUI.
- Engineered Apple Pay + Stripe Connect + Plaid checkout (Stripe PaymentSheet) over a 44-function Firebase backend.
- Led the React Native + Expo iOS build to TestFlight: 900+ tests, EAS CI/CD, code signing, and Sentry observability.

Teaching Assistant

Nashville, TN

Vanderbilt University, Department of Computer Science

Jan. 2025 – Dec. 2025

- Held 12 hours/week of office hours for 50+ students on Java fundamentals, debugging, OOP, & algorithm design.
- Graded 10+ coding assignments per cycle via GitHub & Autograder; improved student resubmissions by 30%.
- Supported 8 Software Engineering course teams; advised on design patterns, testing, and full-stack development.

Software Engineering Intern

Washington, DC

Jenner & Block LLP

June 2024 – Aug. 2024

- Led digital transformation initiative for 50+ attorneys, automating workflows to reduce onboarding costs by 40%.
- Configured Cisco Meraki Access Points across office; improved wireless stability & cut IT tickets by 25%.
- Collaborated with cross-functional teams to integrate 5+ enterprise applications, streamlining operations.

PROJECTS

Sentinel - LLM Adversarial Robustness - Python, React, Express, MongoDB

Feb. 2026 – Present

- At Vanderbilt's Institute of National Security: measured LLM refusal decay vs. adversarial context length.
- Red-teamed Claude, DeepSeek-R1, and closed LLMs (300 probes/model) to identify safety vulnerabilities.
- Fit power-law scaling of attack success vs. context fill; built the Python/React/Express/MongoDB eval engine.

Vectr - Pre-Scene Intelligence for EMTs - Swift, Svelte 5, Rust, Tauri, PostGIS [Whitepaper](#)

Jan. 2026 – May 2026

- Built a 4-platform EMS system (iOS, Tauri, Rust/Actix-web + PostGIS) for pre-arrival scene intelligence.
- On-device AI scribe (Parakeet 4.65% WER + LFM2-350M) converting scene audio into NEMSIS ePCR fields.
- DINOv2 model scoring helicopter landing zones; field-validated with Greene County EMS on a 3-day ride-along.

Spevents.live - Real-time Photo Wall - React, Three.js, Tailwind CSS, AWS S3 [Live](#)

Oct. 2024 – Feb. 2025

- Launched serverless platform with React Three Fiber and S3 + CloudFront, serving 200+ photos at 4 events.
- Created REST API on Vercel with Firebase Auth/Firestore and QR-based zero-download uploads.
- Designed 5 displays (Grid, Slideshow, 3D, Parallax) with Framer Motion and Hugging Face moderation.

QuizLITE - Flashcard Desktop App - C++17, SQLite, Qt6, CMake, GTest [GitHub](#)

Sept. 2024

- Built flashcard app with 3 adaptive modes prioritizing low-accuracy questions via spaced repetition.
- Architected SQLite backend with singleton-pattern CRUD operations and per-question accuracy tracking.
- Configured GitHub Actions with 35+ GoogleTest cases across Ubuntu/macOS, clang-format linting.

AWARDS

Amazon Mission Autonomy Hackathon 2025 - 2nd Place (\$15K Prize) - [Pitch](#) | [GitHub](#)

Nashville, TN

HUSH-MESH, an AWS-powered autonomous maritime defense network with RF-silent drones for routing.

Vanderbilt X Ironsite Hackathon 2025 - 3rd Place (\$2.5K Prize) - [Pitch](#) | [GitHub](#)

Nashville, TN

Built an object permanence model to track blocked objects, boosting accuracy from 12% to 94%.

Google Solutions Challenge 2025 - 1st Place - [Pitch](#) | [GitHub](#)

New York, NY

Gemicast, a real-time outage-risk dashboard + predictive alerts, inspired by recent Tennessee floods.